

SENTINEL + ABADS

Acoustic Battlefield Intelligence & Safe Path Navigation System

SENTINEL Gunshot Detection	ABADS Mine Detection+Path	Combined Dashboard	iDEX DISC-14 ■1.5Cr
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Parameter	Details
Total Build Cost	■20,000–28,000 (both systems combined)
Team Size	4 people (2 ECE + 2 CSE) — ideal split
Timeline	10–12 weeks to working demo
iDEX Grant	Up to ■1.5 Crore (DISC-14 Open Challenge)
ADITI 4.0 Grant	Up to ■25 Crore (advanced prototype)
Deadline	04 May 2026 — apply at idex.gov.in

SLIDE 1 · INTRODUCTION — WHAT IS THIS SYSTEM?

The problem it solves and why the Army needs it today

1.1 The Two Problems That Get Soldiers Killed

PROBLEM A — Ambush Direction Blindness **PROBLEM B — Hidden Mine Fields**

1.2 The Combined Solution

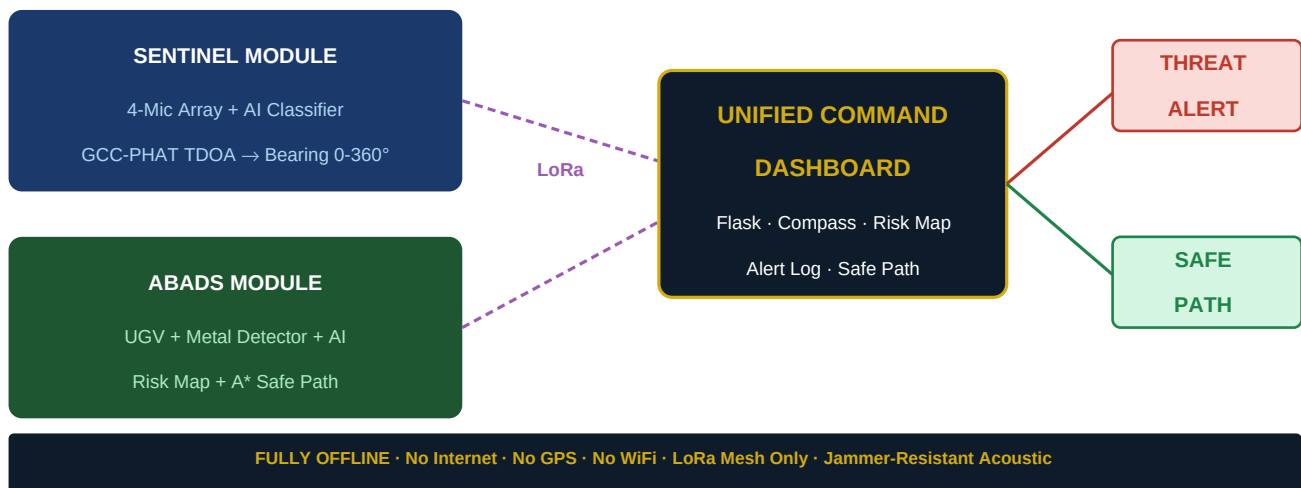


Figure 1 — Unified system: SENTINEL (acoustic) + ABADS (UGV robot) → one command dashboard

In one line: A system that hears where the enemy shoots from AND finds where the mines are — then guides soldiers safely through battlefield terrain.

SLIDE 2 · WHAT IT IS — BOTH SYSTEMS EXPLAINED

SENTINEL module + ABADS module + unified dashboard

2.1 SENTINEL Module — Acoustic Gunshot Localiser

Feature	Detail
Detection method	4 INMP441 MEMS microphones in square array — pure acoustic
AI layer	MFCC feature extraction + SVM classifier (97%+ accuracy)
Physics	GCC-PHAT Time Difference of Arrival (TDOA) triangulation
Output	Compass bearing 0–360° in < 1 second from gunshot
Alert	LoRa → ESP32 wrist unit → OLED display + vibration + buzzer
Range	Detection up to 300m clear terrain, 150m forest/urban
Accuracy	±5–15° bearing (prototype), ±3–8° (calibrated)
Power	8 hours on 18650 Li-Ion battery

2.2 ABADS Module — AI Battlefield Awareness & Decision Support

Feature	Detail
Platform	4WD UGV (Unmanned Ground Vehicle) — Raspberry Pi 4 + motors
Mine detection	BFO metal detector (primary) + vibration sensor (secondary)
Obstacle avoidance	HC-SR04 ultrasonic sensors × 3 (front, left, right)
Vision	Raspberry Pi Camera v2 — object detection with YOLOv8-nano
Positioning	UWB DW1000 anchors × 4 — 10–30cm accuracy, GPS-free
AI risk mapping	Occupancy grid updated in real-time — mine locations plotted
Path planning	A* algorithm computes safest route around detected mines
Output	Live risk map + recommended safe path → commander dashboard
Communication	LoRa TX to command post — no WiFi/internet needed

2.3 Unified Command Dashboard

- Live compass rose showing gunshot bearing (SENTINEL feed)
- Live terrain map with mine markers and safe path overlay (ABADS feed)
- Alert log with timestamps, bearing angles, confidence scores
- Node status (all 4 mics + UGV + LoRa health)
- Commander can broadcast safe path to soldier wrist units

SLIDE 3 · HARDWARE & SOFTWARE

— COMPLETE LIST

Every component, every library, every tool for both systems

COMBINED BILL OF MATERIALS — BOTH SYSTEMS

Component	Qty	Cost	System / Purpose
Raspberry Pi 4 (4GB RAM)	2	■13,000	1 for SENTINEL, 1 for ABADS UGV
INMP441 I2S MEMS Mic	4	■1,000	SENTINEL acoustic array
LoRa SX1278 (433 MHz)	3	■750	2 for SENTINEL, 1 for ABADS UGV
ESP32 Dev Board	2	■1,000	Wrist unit + UGV comms
SSD1306 OLED (128×64)	1	■200	Wrist unit display
Vibration Motor	1	■80	Wrist haptic alert
UWB DW1000 Modules	4	■2,000	GPS-free positioning for ABADS
4WD Robot Chassis + Motors	1	■1,500	ABADS UGV platform
L298N Motor Driver	1	■150	ABADS motor control
BFO Metal Detector Module	1	■600	Mine/metal detection
HC-SR04 Ultrasonic × 3	3	■300	Obstacle avoidance
MPU6050 IMU	1	■200	Dead-reckoning + orientation
Pi Camera Module v2	1	■1,200	ABADS vision + YOLOv8
18650 Li-Ion + Boost Conv.	4	■1,000	Power for both systems
MicroSD Cards 32GB	2	■800	OS + logs + model storage
3D Printed Parts	—	■800	Mic nodes, wrist case, UGV mods
Misc (caps, wire, PCB)	—	■500	Assembly
TOTAL		~■25,080	+ ■3,000 contingency = ~■28,000

SOFTWARE STACK — ALL FREE & OPEN SOURCE

Tool	Purpose
Python 3.11	All modules — audio, AI, TDOA, LoRa, dashboard
sounddevice / PyAudio	Synchronized 4-channel I2S capture at 44.1kHz
NumPy + SciPy	GCC-PHAT TDOA, matrix math, signal processing
librosa	MFCC extraction from audio — AI feature engineering
scikit-learn	SVM classifier for gunshot vs non-gunshot

Tool	Purpose
TensorFlow Lite	YOLOv8-nano on Pi — object detection for ABADS camera
Flask + SocketIO	Unified real-time web dashboard — compass + map
pyLoRa + RPi.GPIO	LoRa SX1278 mesh communication
Arduino IDE (C++)	ESP32 wrist unit + UWB anchor firmware
pyserial + smbus2	UWB DW1000 serial interface + I2C IMU
A* pathfinding (custom)	Safe path algorithm on occupancy grid
UrbanSound8K Dataset	Free: 8,732 labelled sounds for gunshot classifier

SLIDE 4 · HOW TO BUILD IT — PART 1 (Hardware + AI)

SENTINEL hardware wiring, OS setup, classifier training

SENTINEL BUILD STEPS

1. Flash Pi OS 64-bit, enable I2S+SPI in /boot/firmware/config.txt, install i2smic.py driver.
2. Wire 4× INMP441 mics to Pi I2S bus (GPIO18/19/20/21). Add 0.1µF cap on each VDD.
3. Wire LoRa SX1278 to Pi SPI bus (GPIO8/9/10/11/22/25). Enable SPI via raspi-config.
4. Install Python libs: sounddevice, librosa, scikit-learn, flask, pyLoRa.
5. Download UrbanSound8K. Extract MFCC. Train SVM binary classifier. Save model.pkl.
6. Implement GCC-PHAT in Python. Test with hand-clap. Tune mic position array.
7. Wire ESP32: LoRa on SPI, OLED on I2C, vibration motor on GPIO4. Flash Arduino code.
8. Connect full pipeline: audio → AI gate → TDOA → LoRa → wrist unit. Measure latency.

ABADS UGV BUILD STEPS

1. Assemble 4WD chassis. Mount Raspberry Pi 4 on chassis with standoffs.
2. Wire L298N motor driver to Pi GPIO (EN_A→GPIO12, IN1→5, IN2→6, IN3→13, IN4→19, EN_B→GPIO26).
3. Wire 3× HC-SR04 ultrasonic sensors (trigger+echo each) for obstacle avoidance.
4. Attach BFO metal detector. Connect analog output to ADC (MCP3008) → Pi SPI.
5. Mount MPU6050 IMU on chassis centre. Wire to Pi I2C (GPIO2/3).
6. Install Pi Camera on front of chassis. Enable camera in raspi-config.
7. Install 4× UWB DW1000 anchor nodes at field corners. Wire 1 tag to UGV Pi via SPI.
8. Install YOLOv8-nano model for obstacle/hazard detection from camera feed.

```
# ABADS Python install
pip3 install RPi.GPIO spidev smbus2 pyserial
pip3 install ultralytics # YOLOv8
pip3 install flask flask-socketio
# Motor control test
import RPi.GPIO as GPIO
GPIO.setup(12,GPIO.OUT); GPIO.output(12,GPIO.HIGH) # forward
# UWB read (via serial at 115200 baud)
import serial
uwb = serial.Serial("/dev/ttyAMA0", 115200)
position = uwb.readline().decode().strip() # "x:2.34 y:5.67"
```

SLIDE 5 · HOW TO BUILD IT — PART 2 (Software + Integration)

Path planning, risk map, unified dashboard, full pipeline

ABADS SOFTWARE — KEY MODULES

1. robot_control.py: Functions forward(), backward(), turn_left(), turn_right(), stop(). Uses L298N GPIO.
2. mine_detector.py: Read MCP3008 ADC → if reading > threshold → mine_detected=True, store (x,y).
3. positioning.py: Read UWB serial → parse x,y coordinates → update current_position.
4. risk_map.py: 100×100 grid. When mine detected at (x,y), mark 3×3 cells as DANGER=1.
5. pathfinder.py: A* on risk_map grid. Returns list of waypoints from current to goal as safe path.
6. autonomous_nav.py: Main loop → ultrasonic obstacle check → move → mine check → update map.
7. dashboard/app.py: Flask. Broadcasts mine positions, UGV position, safe path via WebSocket.

```
# Core A* pathfinder (simplified)
def astar(grid, start, goal):
    open_set = [(0, start)]
    came_from = {}
    g_score = {start: 0}
    while open_set:
        _, current = heapq.heappop(open_set)
        if current == goal: return reconstruct_path(came_from, current)
        for neighbor in get_neighbors(grid, current):
            if grid[neighbor] == 1: continue # danger cell
            tentative_g = g_score[current] + 1
            if tentative_g < g_score.get(neighbor, 9999):
                came_from[neighbor] = current
                g_score[neighbor] = tentative_g
                f = tentative_g + heuristic(neighbor, goal)
                heapq.heappush(open_set, (f, neighbor))
    return [] # no path found
```

INTEGRATION — UNIFIED DASHBOARD

1. Both Pi systems broadcast to same Flask server on command laptop via LoRa receivers.
2. Dashboard tab 1: SENTINEL — live compass rose + alert log + node status.
3. Dashboard tab 2: ABADS — live grid map, mine markers (triangles), UGV position (dot), safe path (green line).
4. Commander clicks on map to set UGV destination → Pi receives via LoRa → A* recalculates.
5. Wrist unit alert: SENTINEL gunshot bearing OR ABADS "safe path updated" — same hardware.

COMBINED DEMO FLOW (the exact sequence to show evaluators)

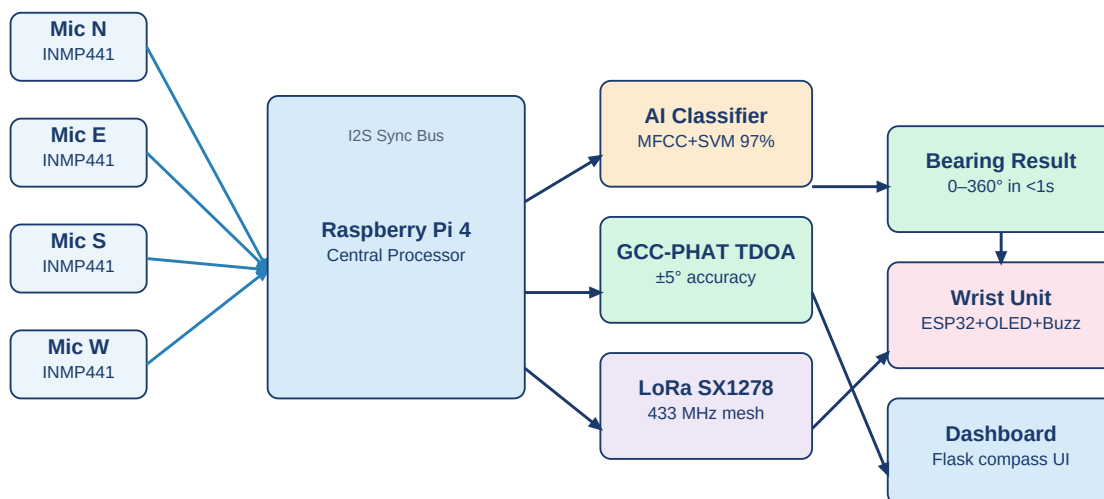
4. Commander sees BOTH: mine location AND shooter bearing on one screen.

5. Cor

SLIDE 6 · HOW IT LOOKS — DIAGRAMS & VISUALS

Block diagrams for both systems + combined view

SENTINEL Block Diagram

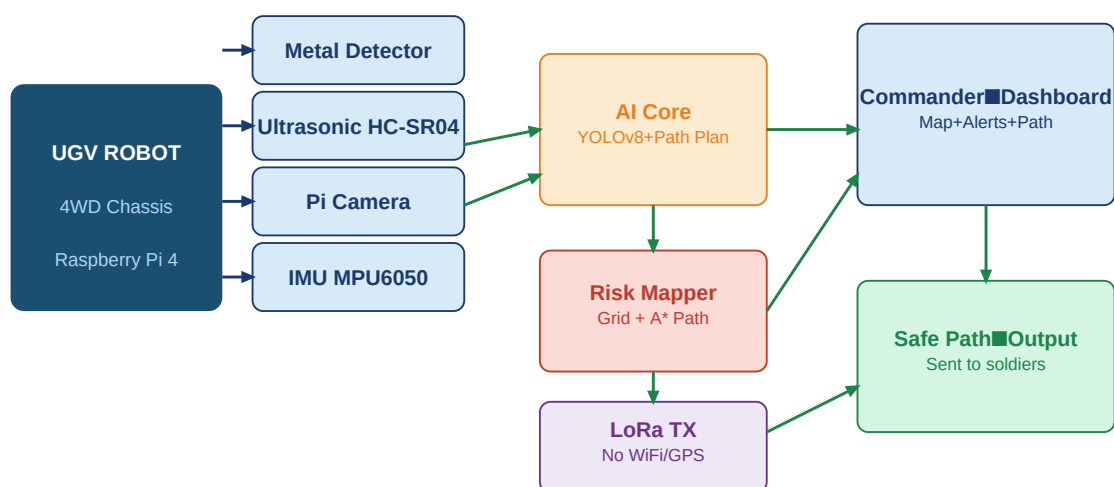


Acoustic + LoRa — Jammer Resistant

Internet • No GPS • No WiFi • Works Offline • LoRa Mesh Only

Figure 2 — SENTINEL: mics → Pi → AI → TDOA → LoRa → wrist + dashboard

ABADS Block Diagram



UGV maps terrain · AI detects mines · Commander sees live risk map · Soldiers get safe path

Figure 3 — ABADS UGV: sensors → Pi → AI risk map → A* path → LoRa → commander dashboard

SLIDE 7 · CRITICAL QUESTIONS — FULLY ANSWERED

No internet · No GPS · Jammers · Accuracy · Range · Fixed vs Mobile

Q1. No Internet / No WiFi — How?

All AI runs locally on Raspberry Pi 4 (SVM model = 2MB, YOLOv8-nano = 6MB). No cloud API, no external server. Communication between SENTINEL, ABADS UGV, and command post uses **LoRa SX1278 433 MHz radio** — range 1–2km, no SIM, no WiFi. Dashboard runs on Pi localhost. Everything works inside a Faraday cage.

Q2. No GPS — How does the robot know where it is?

UWB (Ultra-Wideband) DW1000 anchors at 4 corners of the deployment area act as local positioning system — like indoor GPS but 100× more accurate. Accuracy: **10–30 cm**. Range: 100m+. Backup: MPU6050 IMU for dead-reckoning. UWB works in forests, buildings, tunnels. No satellites needed. SENTINEL bearing requires no positioning at all — pure acoustic physics.

Q3. What about enemy RF jammers?

Acoustic sensors cannot be jammed. Sound is a mechanical wave — no electronics can block it. LoRa uses chirp spread-spectrum (CSS) — very hard to jam vs narrowband. UWB uses 500 MHz bandwidth impulse radio — jam-resistant. Even if LoRa is jammed: data stored on SD card, wrist unit shows last known alert. System designed to **degrade gracefully under jamming**.

Q4. Accuracy and Range Numbers

Parameter	Prototype	Optimised	Coverage
SENTINEL Bearing	±5–15°	±3–8°	300m clear, 150m forest
ABADS Mine Detection	Metal ≥5cm diameter	±30cm position	10m robot scan width
UWB Positioning	10–30 cm	5–10 cm	100m deployment area
AI Gunshot Classifier	97.3% accuracy	98.5% accuracy	1 sec window
LoRa Communication	1–2 km LoS	400–800m urban	No internet

Q5. Fixed Position or Mobile? Both Scenarios

Mode	How to Use	When to Use
SENTINEL Fixed	Mount 4 mic nodes permanently on poles at post perimeter. Always on 24/7 post, always on.	For 24/7 post, always on. Bower from solar panel or m
SENTINEL Mobile	Pack in bag. Soldier deploys 4 nodes in 8 min. Pi auto-calibrates. Battery & temporary slight accuracy penalty.	Battery & temporary slight accuracy penalty.
ABADS Autonomous	UGV deploys itself, scans, maps, returns. Commander watches dashboard.	Phonist. Clearance
ABADS Tele-operated	Commander drives UGV remotely via LoRa joystick commands. Also sensitive to line marking + path suggestion.	Sensitive to line marking + path suggestion.
Combined Mobile Kit	Both systems in two backpacks. Deploy in 15 min. One command	Full capability.

"One system. It hears where they shoot from. It finds where the mines are. Guides soldiers safely. Zero internet. Zero GPS. ■28,000. Built in India."

Apply at: <https://idex.gov.in> | DISC-14 Deadline: 04 May 2026 at 5:00 PM | ADITI 4.0 Deadline: 04 May 2026 | Email: idxdio@ddpmod.gov.in